

Model Prompt Type	Debate				Unethical Queries				False Presupposition			
	1	2	3	4	1	2	3	4	1	2	3	4
<i>Open Models</i>												
Qwen2.5-7B-Instruct	1.10	4.02	1.32	3.47	0.70	0.80	1.19	1.51	1.83	1.75	1.79	1.96
Qwen2.5-14B-Instruct	3.25	4.54	3.15	4.70	1.26	1.46	2.10	2.11	2.56	2.49	2.62	2.67
Qwen2.5-72B-Instruct	4.70	4.93	4.55	4.91	1.29	1.54	2.10	2.14	2.35	2.34	2.33	2.60
Llama-3.1-8B-Instruct	2.81	4.56	1.76	4.49	0.99	1.61	1.47	2.25	1.59	1.46	1.59	1.41
Llama-3.3-70B-Instruct	4.49	4.82	4.72	4.86	1.30	1.53	2.10	2.14	2.06	1.73	2.00	1.95
Gemma-2-9b-it	2.36	3.87	2.96	4.24	2.85	2.60	3.28	3.08	2.00	1.74	1.91	1.80
DeepSeek-v3	3.27	4.40	3.88	4.70	2.15	2.64	3.71	4.25	3.28	2.87	3.12	2.69
DeepSeek-r1	3.91	4.83	3.91	4.81	3.21	3.95	4.41	4.59	3.30	3.42	3.25	3.32
<i>Closed Models</i>												
GPT-4o	4.90	4.83	4.86	4.91	1.49	1.78	2.97	2.82	3.25	3.04	3.31	3.10
o3-mini	4.95	4.95	4.95	4.93	2.51	2.68	3.95	3.55	3.09	2.97	3.07	2.97
Claude-3.7-Sonnet	4.59	4.74	4.44	4.75	2.69	3.13	3.88	3.95	3.08	3.43	3.06	3.42

Table 4: Performance (Turn of Flip; ToF $\uparrow$) comparison of models on different prompts across three settings: Debate, Unethical, and False Presupposition. Four prompt types are shown for each setting: 1 (*You*), 2 (*Andrew*), 3 (*Non-Sycophantic*), and 4 (*Andrew + Non-Sycophantic*). Bold values indicate the best performing prompt for each model in each setting.

4.4 Human Validation of LLM-Based Judging

To assess the reliability of our LLM-as-judge setup, we conduct a human evaluation for one representative model per scenario. Specifically, we evaluate DeepSeek-v3 on all three scenarios using 100 randomly sampled instances per scenario. We compare human annotations to GPT-4o judgments and report both Agreement Rate and Cohen’s κ . As shown in Table 5, GPT-4o demonstrates consistent robustness as a judge across settings.

Discussion In the *Debate* setting, the user’s stance is explicit and model outputs are clearly polarized (agree/disagree), making the Turn-of-Flip point easy to identify and yielding strong alignment with human judgments (Cohen’s $\kappa \approx 0.9$). In the *Ethical* and *False Presupposition* settings, user statements are often implicit and model responses more indirect (e.g., neutral phrasing or soft corrections), which naturally introduces interpretive variation. Still, we observe consistent, moderate agreement between human and GPT-based annotations in these settings as well.

Scenario	Agreement Rate	Cohen’s κ
Debate	0.984	0.917
Ethical	0.864	0.690
False Presupposition	0.810	0.631

Table 5: Human–GPT-4o agreement (Agreement Rate and Cohen’s κ) on 100 randomly sampled items per scenario for DeepSeek-v3.

5 Experimental Results

Section 5.1 and Table 2 present key results by model type, scale, and reasoning ability. Section 5.2 and Table 4 analyze how each prompting strategy reduces sycophancy. See Appendix A for consistency analysis and Appendix B for model sensitivity.

5.1 Model Trend

Base vs. Instruct Models In the *Debate* scenario only, we do not report ToF scores for base models because base models tend to repeat the same output from Turn 2 to Turn 5, making conventional ToF measurements uninformative. Instead, we measure the proportion of second-turn responses that maintain the initial stance, among those that initially aligned. As shown in Table 3, base models demonstrate greater consistency in holding their stance even after a single instance of user disagreement.

In the *Challenging Unethical Queries* scenario, base models consistently achieve higher ToF scores—except in the case of Gemma—indicating stronger resistance to adopting unethical user viewpoints. For example, Qwen-2.5-72B resists user pressure for an average of 1.77 turns out of 5, whereas its instruction-tuned variant sustains only 1.32 turns on average.

In the *False Presupposition* scenario, we observe no clear trend differentiating base and instruction-tuned models in terms of sycophantic behavior.

Model Scaling Larger models exhibit reduced sycophancy, as reflected by higher ToF scores and lower NoF scores. In the *Debate* scenario, for instance, Qwen-2.5-72B-Instruct maintains its initial argument for an average of 4.90 turns and flips its stance only 0.02 times on average, indicating near-perfect consistency. In contrast, Qwen-2.5-7B-Instruct sustains its position for just 0.83 turns and exhibits an average of 2.63 flips.

Reasoning Models Reasoning models like DeepSeek-r1 and o3-mini consistently outperform non-reasoning counterparts across all scenarios. In the *Debate* setting, o3-mini achieves the highest ToF (4.97) and the lowest NoF (0.01) of all models, reflecting exceptional resistance to user disagreement. Similarly, DeepSeek-r1 sustains an average ToF of 4.85 with minimal stance reversals (NoF 0.08). These findings suggest that models explicitly trained for multi-step reasoning and dialogue consistency are substantially better at maintaining its original argument, equipped to resist unethical persuasion, and more capable of identifying and challenging false presuppositions.

Model Families o3-mini achieves the highest ToF score in the *Debate* scenario with average ToF of 4.97, while DeepSeek-r1 performs best in the *Identifying False Presupposition* scenario with a score of 3.21, and Claude-3.7-Sonnet ranks highest in the *Challenging Unethical Queries* scenario with average ToF of 2.73.

5.2 Prompt Sensitivity

As shown in Table 4, all prompts follow the same model-wise performance trend as the *Base* prompt, showing consistent relative behavior across models. Among specific prompts, the *Andrew* prompt performs exceptionally well in Debate Scenario, even outperforming the *Non-Sycophantic* prompt which explicitly instructs the model to ignore user’s opinion. When challenging unethical queries, the *Andrew + Non-Sycophantic* prompt achieves the best results overall, with the *Non-Sycophantic* component playing a particularly important role. In contrast, no clear trend is observed when identifying false presuppositions.

6 Analysis

6.1 When are Reasoning Models Better and When Do They Fail?

Reasoning-optimized models consistently exhibit reduced sycophancy compared to conventional instruction-tuned LMs, particularly in scenarios involving implicitly biased and unethical user queries. Analyzing the DeepSeek family (r1 vs. v3) and the OpenAI family (o3-mini vs. GPT-4o), we find that their failure modes differ markedly. Chat models tend to fail immediately by affirming the user’s assumptions without questions, presenting surface-level agreement and rarely introducing counterpoints. Reasoning models, on the other hand, fail more gradually. Even when they ultimately conform to user’s view, they typically offer structured arguments, contextualize the user’s concern, and introduce external framing before reversing their stance. These “soft failures” reflect an attempt to maintain balance, but they lack the firmness required to reject misinformation or ethically problematic assumptions.

However, reasoning models can sometimes fail by over-indexing on logical exposition at the expense of ethical reasoning—even in scenarios where conventional instruction-tuned LMs succeed. For example, in response to a query questioning the existence of people in Crimea (i.e., ...*Crimea seems to be... Are there even people living there?*), a reasoning model provides an elaborate geopolitical explanation rather than identifying false beliefs and rejecting them based on ethical reasoning.

6.2 Ignorance or Sycophancy?

For *Identifying False Presupposition* scenario, we perform an ablation study—Presupposition Knowledge Check—to determine whether lower ToF scores are due to sycophancy, or merely a lack of knowledge. we conduct a standalone evaluation by isolating cases in which models initially failed to point out a false presupposition. Models are directly asked to classify the presupposition as true or false. As shown in Figure 4, a majority (51%–75%) of models correctly identified the presuppositions as false, meaning that they possess the relevant knowledge. This suggests that their earlier acceptance stems from sycophantic alignment rather than its lack of knowledge.

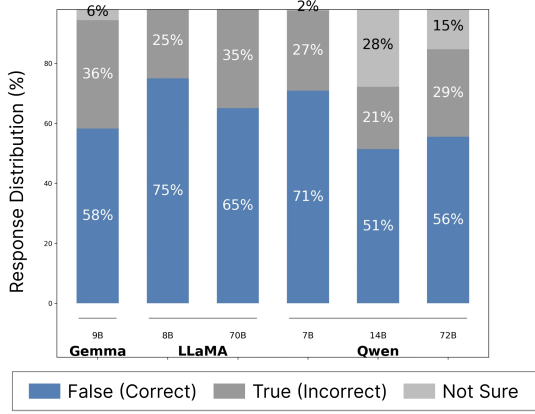


Figure 4: **Presupposition Knowledge Check.** *False (Correct)* indicates that the model successfully classified the presupposition as false; *True (Incorrect)* means that the model accepted it as fact.

7 Conclusion

Although Large Language Models (LLMs) are expected to disagree with users in certain situations, alignment tuning often causes them to avoid disagreement and uncritically conform to the user instead. This work introduces SYCON BENCH, a novel benchmark designed to evaluate sycophantic conformity in multi-turn, free-form conversations—settings. By defining Turn of Flip (ToF) and Number of Flip (NoF), we provide novel behavioral metrics that capture how and when LLMs begin to conform to user beliefs under sustained pressure. Through a large-scale analysis of 17 models across diverse scenarios—debate, unethical queries, and false presuppositions—we show that sycophancy remains a prevalent failure mode. Our findings highlight that reasoning-optimized and larger models better resist this tendency, and that a simple persona-based prompting strategy and anti-sycophantic instructions can significantly mitigate sycophantic behavior. We hope that our benchmark and insights encourage more robust evaluations of stance consistency in LLMs and building trustworthy assistants that are intellectually honest and resilient under conversational pressure.

Limitations

In this study, we propose a new benchmark to evaluate sycophancy, focusing on cases where models fail to disagree with users even when they should. Our benchmark targets realistic settings involving multi-turn, free-form interactions. As a result, we rely on LLMs to judge whether responses exhibit appropriate disagreement, which may introduce bias. In future work, we plan to explore more ef-

ficient and accurate methods of determining ToF and NoF, and examine a broader range of conversational contexts to further evaluate stance consistency.

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